Hanyang Chen

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Education

Beijing Jiaotong University **Beijing, China**
M.S. in Computer Science and Technology *Sep. 2023 - Jun. 2026*

- **Advisor:** Shengnan Guo and Huaiyu Wan; **GPA:** 3.58/4.00
- **Research Interests:** Reinforcement Learning, Data Mining, Large Language Models

Beijing Jiaotong University **Beijing, China**
B.S. in Computer Science and Technology *Sep. 2019 - Jun. 2023*

- **GPA:** 3.83/4.00

Publications and Preprints

Publications

- **DADiff: Diffusion-Driven Cross-Domain Policy Adaptation for Reinforcement Learning**
  Hanyang Chen, Anirudh Satheesh, Longchao Da, Hua Wei
  The 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. (Accepted)
- **Online Irregular Multivariate Time Series Forecasting via Uncertainty-Driven Dual-Expert Calibration**
  Haonan Wen, **Hanyang Chen**, Songhe Feng
  The 32nd SIGKDD Conference on Knowledge Discovery and Data Mining, 2026. (Accepted)
- **Proactive-XLight: Proactive Traffic Signal Control with Pluggable and Reliable Traffic Prediction**
  Yang Jiang, Shengnan Guo, **Hanyang Chen**, Xiaowei Mao, Youfang Lin, Huaiyu Wan
  IEEE Transactions on Mobile Computing, 2025. (IF=9.2)
- **DiffLight: A Partial Rewards Conditioned Diffusion Model for Traffic Signal Control with Missing Data**
  Hanyang Chen, Yang Jiang, Shengnan Guo, Xiaowei Mao, Youfang Lin, Huaiyu Wan
  The 38th Annual Conference on Neural Information Processing Systems, 2024. (Spotlight, Top 3%)

Preprints

- **Beyond English: Uncovering the Multilingual Gap in Vision-Language-Action Models**
  Hanyang Chen, Hongliang Li, Jiarui Cao, Yang Li, Yang Jiang, Haonan Wen, Kaiyu Huang, Shengnan Guo, Huaiyu Wan
  arXiv:2606.15714, 2026.

Services

- **Teaching Assistant:** C402002B, Deep Learning, Fall 2024 and Fall 2025
- **Reviewer:** Data Intelligence, CCKS 2025, NeurIPS 2025–2026, WACV 2026–2027, KDD 2026

Patents

- **A Traffic Signal Control Method Based on DRL and Situation Prediction**, CN202410357941.4
  Huaiyu Wan, Shengnan Guo, Yang Jiang, **Hanyang Chen**, Jingwen Chen, Youfang Lin

Competitions

- **National E-Commerce Challenge Competition for College Students** *Jun. 2025*
  Team Member, Provincial 1st Prize
- **National College Student Computer System Capability Competition** *Aug. 2022*
  Team Member, National 3rd Prize
- **MathorCup Mathematical Modeling Challenge** *Jun. 2022*
  Team Member, National 2nd Prize

Awards and Scholarships

- **Outstanding Graduate Students of Beijing Jiaotong University**, Beijing Jiaotong University, 2026
- **The First Prize Academic Scholarship**, Beijing Jiaotong University, 2024, 2025
- **National Scholarship**, Beijing Jiaotong University, 2024
- **NeurIPS 2024 Scholar Award**, NeurIPS Committee, 2024
- **Outstanding Graduate Students of Beijing**, Beijing Jiaotong University, 2023

Research Experience

Multilingual Evaluation for Vision-Language-Action Models *Mar. 2026 - May. 2026*

Project Leader, Advisor: Prof. Shengnan Guo

- This project focuses on evaluating multilingual robustness in Vision-Language-Action models, where robotic agents must understand and execute instructions across different languages and code-switching scenarios.
- It studies how language inputs affect embodied policy learning, revealing a significant multilingual performance gap in VLAs driven by representation shifts and misalignment between language understanding and action execution.
- It proposes Multilingual Principal Component Alignment, a PCA-based representation alignment method that reduces multilingual gaps and improves cross-lingual robustness on robotic benchmarks.

Online Irregular Time Series Forecasting with Uncertainty Estimation *Nov. 2025 - Feb. 2026*

Project Member, Advisor: Prof. Songhe Feng

- Instead of relying on temporal continuity, the framework uses uncertainty estimation as a signal to detect distribution shifts and guide online adaptation decisions.
- A dual-expert calibration design is introduced, separating reliable and unreliable samples to enable stable adaptation while avoiding interference from high-uncertainty data.
- The method achieves efficient, model-agnostic online learning by freezing the original forecaster and updating only lightweight calibration modules for real-world dynamic environments.

Cross-Domain Policy Adaptation via Generative Trajectory Mismatch *Jun. 2025 - Sep. 2025*

Project Leader, Advisor: Prof. Hua Wei

- This work addresses sim-to-real reinforcement learning under limited target interactions by modeling source–target dynamics mismatch as a fine-grained distributional discrepancy.
- Instead of coarse domain alignment, the framework introduces a diffusion-based generative trajectory view of state transitions through a theoretically grounded performance bound.
- Multiple experimental results and discussions demonstrate the effectiveness of our method and further explore the properties of our method.

Cross-City Traffic Signal Control from Latent Space Perspective *Nov. 2024 - May. 2025*

Project Leader, Advisor: Prof. Shengnan Guo

- This project focuses on multi-scenario traffic signal control, aiming to improve generalization and collaboration across different urban traffic environments through joint training on multiple city scenarios.
- To alleviate scenario-level negative transfer, the framework introduces a Shapley-guided Mixture of Experts that adaptively routes diverse traffic patterns to specialized experts for more robust representations.
- The framework further designs a Mutual Decision Distillation Block to align latent decision estimations across neighboring agents, achieving more stable collaboration and superior performance in both multi-scenario and zero-shot traffic control tasks.

Traffic Signal Control under Data Missing Scenarios *Nov. 2023 - May. 2024*

Project Leader, Advisor: Prof. Shengnan Guo

- This project focuses on traffic signal control under missing-data scenarios, improving urban traffic efficiency when sensor failures or incomplete observations occur in real-world traffic networks.
- We develop an offline diffusion-model-based framework that jointly performs traffic state imputation and traffic signal control through conditional generative modeling.
- The framework integrates partial-reward conditioned diffusion, spatial-temporal transformer modeling, and diffusion communication, enabling robust decision-making under diverse missing-data scenarios.

Skills

Programming	Python (numpy, pandas, matplotlib, pytorch), C, C++
Languages	Chinese (native), English
Software	Latex, Markdown, Notion, Excel